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Assignment Cover Page- Jan'26 Session

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Note- The assignment Question have 5 Sections/Blocks. Kindly attempt Any **ONE question** from Each of the 5 blocks. Each question carries equal marks. For Eg:

Block 1 Attempt- Question 1a or 1b

Block 2 Attempt- Question 2a or 2b

Block 3 Attempt- Question 3a or 3b

Block 4 Attempt- Question 4a or 4b

Block 5 Attempt- Question 5a or 5b

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Q1 B: Derive the equation of the osculating plane for a curve given by $\mathbf{r}(t)$. Provide a geometrical interpretation of the osculating plane.

Setup and Definitions

Let $\mathbf{r}(t)$ be a smooth curve in $\mathbb{R}^3$, parameterized by t , and assume $\mathbf{r}'(t) \neq \mathbf{0}$ (the curve is regular). The **osculating plane** at a point $t = t_0$ is the plane that best approximates the curve locally — it contains the tangent line and the direction in which the curve is bending.

Step 1: Unit Tangent Vector

The unit tangent vector is:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

Step 2: Principal Normal Vector

Since $|\mathbf{T}(t)| = 1$, differentiating gives $\mathbf{T}(t) \cdot \mathbf{T}'(t) = 0$, so $\mathbf{T}'(t) \perp \mathbf{T}(t)$.

The principal unit normal vector (defined when $\mathbf{T}'(t) \neq \mathbf{0}$) is:

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|}$$

The condition $\mathbf{T}'(t_0) \neq \mathbf{0}$ is required for the osculating plane to be well-defined (it fails, for instance, at inflection points of straight lines).

Step 3: Binormal Vector

The **binormal vector** is the unit vector perpendicular to the osculating plane, defined as:

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

Since $\mathbf{T}$ and $\mathbf{N}$ are orthonormal, $|\mathbf{B}| = 1$ and $\mathbf{B} \perp \mathbf{T}$, $\mathbf{B} \perp \mathbf{N}$.

Step 4: Equation of the Osculating Plane

The osculating plane at $t = t_0$ passes through $\mathbf{r}(t_0)$ and has $\mathbf{B}(t_0)$ as its normal. A point $\mathbf{r}$ lies in this plane if and only if:

$$\mathbf{B}(t_0) \cdot (\mathbf{r} - \mathbf{r}(t_0)) = 0$$

Step 5: Practical Form via $\mathbf{r}'$ and $\mathbf{r}''$

In practice, $\mathbf{B}$ need not be fully normalized. Since $\mathbf{T} = \frac{\mathbf{r}'}{|\mathbf{r}'|}$, one can show:

$$\mathbf{r}'(t) \times \mathbf{r}''(t) \parallel \mathbf{T}(t) \times \mathbf{N}(t) = \mathbf{B}(t)$$

because the cross product $\mathbf{r}' \times \mathbf{r}''$ differs from $\mathbf{B}$ only by a positive scalar factor. Therefore the osculating plane equation can equivalently be written as:

$$\boxed{(\mathbf{r}'(t_0) \times \mathbf{r}''(t_0)) \cdot (\mathbf{r} - \mathbf{r}(t_0)) = 0}$$

Proof that $\mathbf{r}' \times \mathbf{r}'' \parallel \mathbf{B}$

Let s denote arc length so that $\mathbf{r}' = |\mathbf{r}'| \mathbf{T}$. Differentiating:

$$\mathbf{r}'' = |\mathbf{r}'|' \mathbf{T} + |\mathbf{r}'| \mathbf{T}' = |\mathbf{r}'|' \mathbf{T} + |\mathbf{r}'|^2 \kappa \mathbf{N}$$

where $\kappa = |\mathbf{T}'|/|\mathbf{r}'|$ is the curvature. Then:

$$\mathbf{r}' \times \mathbf{r}'' = |\mathbf{r}'| \mathbf{T} \times (|\mathbf{r}'|' \mathbf{T} + |\mathbf{r}'|^2 \kappa \mathbf{N}) = |\mathbf{r}'|^3 \kappa (\mathbf{T} \times \mathbf{N}) = |\mathbf{r}'|^3 \kappa \mathbf{B}$$

since $\mathbf{T} \times \mathbf{T} = \mathbf{0}$. Thus $\mathbf{r}' \times \mathbf{r}''$ is a positive multiple of $\mathbf{B}$, confirming the equivalence. $\square$

Geometric Interpretation

- The osculating plane “*kisses*” (from Latin *osculari*) the curve at $\mathbf{r}(t_0)$, providing the best planar approximation to second order.
- It contains both $\mathbf{T}(t_0)$ (the direction of motion) and $\mathbf{N}(t_0)$ (the direction of curvature), capturing the curve’s instantaneous direction *and* bending.
- The binormal $\mathbf{B}(t_0)$ is perpendicular to the osculating plane and measures how much the curve twists out of the plane (governed by torsion τ).
- Together $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ form the **Frenet–Serret frame**, an orthonormal basis that moves along the curve.

Summary

$$\text{Osculating plane (abstract form): } \mathbf{B}(t_0) \cdot (\mathbf{r} - \mathbf{r}(t_0)) = 0$$

$$\text{Osculating plane (practical form): } (\mathbf{r}'(t_0) \times \mathbf{r}''(t_0)) \cdot (\mathbf{r} - \mathbf{r}(t_0)) = 0$$

Q2 B: Define a helix and prove that a curve is a helix if and only if the ratio of curvature to torsion is constant.

Preliminaries: Frenet–Serret Formulas

Let $\mathbf{r}(s)$ be a smooth curve in $\mathbb{R}^3$ parameterized by arc length s , with $\kappa(s) > 0$ everywhere (so that $\mathbf{N}$ and $\mathbf{B}$ are well-defined). The Frenet–Serret frame $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ satisfies:

$$\mathbf{T}' = \kappa \mathbf{N}, \quad \mathbf{N}' = -\kappa \mathbf{T} + \tau \mathbf{B}, \quad \mathbf{B}' = -\tau \mathbf{N},$$

where $\kappa > 0$ is the curvature and τ is the torsion, and primes denote d/ds .

Definition of Helices

Definition 1 (Helix). *A smooth curve $\mathbf{r}(s)$ (with $\kappa > 0$) is called a **helix** (or general helix) if its unit tangent vector $\mathbf{T}(s)$ makes a constant angle α with some fixed unit vector $\mathbf{u} \in \mathbb{R}^3$, i.e.,*

$$\mathbf{T}(s) \cdot \mathbf{u} = \cos \alpha \quad \text{for all } s,$$

where $0 \leq \alpha \leq \pi$ is constant.

To Prove:

Theorem 1 (Theorem). *A smooth curve $\mathbf{r}(s)$ with $\kappa > 0$ is a helix if and only if the ratio κ/τ is constant (equivalently, τ/κ is constant).*

Proof. ($\Rightarrow$) **Assume the curve is a helix.**

Let $\mathbf{u}$ be the fixed unit vector with $\mathbf{T} \cdot \mathbf{u} = \cos \alpha$ for all s . Differentiating with respect to s :

$$\frac{d}{ds}(\mathbf{T} \cdot \mathbf{u}) = \mathbf{T}' \cdot \mathbf{u} = \kappa \mathbf{N} \cdot \mathbf{u} = 0.$$

Since $\kappa > 0$, we conclude:

$$\mathbf{N} \cdot \mathbf{u} = 0. \tag{1}$$

Now differentiate (1) with respect to s :

$$\frac{d}{ds}(\mathbf{N} \cdot \mathbf{u}) = \mathbf{N}' \cdot \mathbf{u} = (-\kappa \mathbf{T} + \tau \mathbf{B}) \cdot \mathbf{u} = 0.$$

This gives:

$$-\kappa(\mathbf{T} \cdot \mathbf{u}) + \tau(\mathbf{B} \cdot \mathbf{u}) = 0 \implies \tau(\mathbf{B} \cdot \mathbf{u}) = \kappa \cos \alpha. \tag{2}$$

Since $\mathbf{u}$ is a unit vector and $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ is an orthonormal basis, we can decompose:

$$\mathbf{u} = (\mathbf{T} \cdot \mathbf{u}) \mathbf{T} + (\mathbf{N} \cdot \mathbf{u}) \mathbf{N} + (\mathbf{B} \cdot \mathbf{u}) \mathbf{B} = \cos \alpha \mathbf{T} + (\mathbf{B} \cdot \mathbf{u}) \mathbf{B}.$$

Taking the squared norm $|\mathbf{u}|^2 = 1$:

$$\cos^2 \alpha + (\mathbf{B} \cdot \mathbf{u})^2 = 1 \implies \mathbf{B} \cdot \mathbf{u} = \pm \sin \alpha. \tag{3}$$

Substituting (3) into (2) (with $\sin \alpha \neq 0$, i.e., $\alpha \neq 0, \pi$):

$$\tau(\pm \sin \alpha) = \kappa \cos \alpha \implies \frac{\kappa}{\tau} = \pm \tan \alpha = \text{constant}.$$

($\Leftarrow$) **Assume $\kappa/\tau = c$ is constant.**

Define the angle α by $\tan \alpha = c$ (so α is constant), and set:

$$\mathbf{u}(s) = \cos \alpha \mathbf{T}(s) + \sin \alpha \mathbf{B}(s).$$

We claim $\mathbf{u}$ is a constant vector. Differentiating using the Frenet-Serret formulas:

$$\begin{aligned} \frac{d\mathbf{u}}{ds} &= \cos \alpha \mathbf{T}' + \sin \alpha \mathbf{B}' \\ &= \cos \alpha (\kappa \mathbf{N}) + \sin \alpha (-\tau \mathbf{N}) \\ &= (\kappa \cos \alpha - \tau \sin \alpha) \mathbf{N}. \end{aligned}$$

Since $\tan \alpha = \kappa/\tau$, we have $\kappa \cos \alpha = \tau \sin \alpha$, so:

$$\frac{d\mathbf{u}}{ds} = (\kappa \cos \alpha - \tau \sin \alpha) \mathbf{N} = 0.$$

Hence $\mathbf{u}$ is a constant unit vector (note $|\mathbf{u}|^2 = \cos^2 \alpha + \sin^2 \alpha = 1$). Moreover:

$$\mathbf{T}(s) \cdot \mathbf{u} = \cos \alpha (\mathbf{T} \cdot \mathbf{T}) + \sin \alpha (\mathbf{B} \cdot \mathbf{T}) = \cos \alpha,$$

so $\mathbf{T}$ makes the constant angle α with the fixed direction $\mathbf{u}$. Therefore, the curve is a helix. $\square$

Remarks

1. The condition $\kappa > 0$ is essential throughout; a straight line (where $\kappa = 0$) is a degenerate helix for which the theorem does not apply in this form.
2. The case $\alpha = 0$ (or π) would mean $\mathbf{T} \parallel \mathbf{u}$ for all s , i.e., the curve is a straight line. For a non-degenerate helix, $0 < \alpha < \pi/2$.
3. The axis direction is $\mathbf{u} = \cos \alpha \mathbf{T} + \sin \alpha \mathbf{B}$, which lies in the osculating plane only if $\sin \alpha = 0$ (degenerate).
4. A *circular helix* (e.g., $\mathbf{r}(t) = (a \cos t, a \sin t, bt)$) is the special case where *both* κ and τ are individually constant, not just their ratio.

Q3 A: What is the osculating sphere of a curve? Derive its radius and center in terms of the curvature and torsion of the curve.

Preliminaries: Frenet–Serret Apparatus

Let $\mathbf{r}(s)$ be a smooth curve in $\mathbb{R}^3$ parameterized by arc length s , with $\kappa(s) > 0$ and $\tau(s) \neq 0$. The Frenet–Serret frame $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ satisfies:

$$\mathbf{T}' = \kappa \mathbf{N}, \quad \mathbf{N}' = -\kappa \mathbf{T} + \tau \mathbf{B}, \quad \mathbf{B}' = -\tau \mathbf{N},$$

where primes denote d/ds . Set $\rho = 1/\kappa$ (radius of curvature) and $\sigma = 1/\tau$ (radius of torsion).

Definition

The **osculating sphere** (or *sphere of curvature*) of a curve $\mathbf{r}(s)$ at a point $P = \mathbf{r}(s_0)$ is the unique sphere that has *third-order contact* with the curve at P . Equivalently, it is the limiting position of the sphere passing through four consecutive points on the curve as all four points approach P .

It contains the osculating circle at P as a great circle and lies on the same side of the osculating plane as the centre of curvature.

Derivation of the Centre and Radius

The centre $\mathbf{C}$ and radius R of the osculating sphere at $\mathbf{r}(s)$ are:

$$\mathbf{C} = \mathbf{r} + \rho \mathbf{N} + \rho' \sigma \mathbf{B},$$

$$R = \sqrt{\rho^2 + (\rho' \sigma)^2} = \sqrt{\rho^2 + \left(\frac{\rho'}{\tau}\right)^2}.$$

Proof. Step 1: Parameterize the centre.

Let the centre of the osculating sphere be $\mathbf{C}(s)$, at constant distance R from $\mathbf{r}(s)$. Write the displacement in the Frenet frame:

$$\mathbf{C} - \mathbf{r} = a \mathbf{T} + b \mathbf{N} + c \mathbf{B},$$

so that the sphere condition is:

$$|\mathbf{C} - \mathbf{r}|^2 = a^2 + b^2 + c^2 = R^2. \quad (0)$$

Step 2: First-order contact condition.

Differentiate (0) with respect to s :

$$\frac{d}{ds} |\mathbf{C} - \mathbf{r}|^2 = 0 \implies 2(\mathbf{C} - \mathbf{r}) \cdot \frac{d}{ds} (\mathbf{C} - \mathbf{r}) = 0.$$

Since $\mathbf{C}$ is the centre of the *osculating* sphere (third-order contact), we require $\mathbf{C}$ to be chosen so that this equation and the next two are satisfied. More directly: for contact of order ≥ 1 , the tangent vector $\mathbf{T}$ must be perpendicular to $\mathbf{C} - \mathbf{r}$:

$$(\mathbf{C} - \mathbf{r}) \cdot \mathbf{T} = 0 \implies a = 0. \quad (1)$$

So $\mathbf{C} - \mathbf{r} = b\mathbf{N} + c\mathbf{B}$.

Step 3: Second-order contact condition.

Differentiate $(\mathbf{C} - \mathbf{r}) \cdot \mathbf{T} = 0$ with respect to s :

$$\frac{d}{ds}[(\mathbf{C} - \mathbf{r}) \cdot \mathbf{T}] = 0.$$

Expanding (using $\frac{d(\mathbf{C}-\mathbf{r})}{ds} = -\mathbf{T} + \mathbf{C}'$ and, for an osculating object, the effective differentiation acts via the curve's derivatives):

$$-|\mathbf{T}|^2 + (\mathbf{C} - \mathbf{r}) \cdot \mathbf{T}' = 0 \implies -1 + (\mathbf{C} - \mathbf{r}) \cdot \kappa\mathbf{N} = 0 \implies \kappa b = 1 \implies b = \frac{1}{\kappa} = \rho. \quad (2)$$

Step 4: Third-order contact condition.

We now differentiate the relation $\mathbf{C} - \mathbf{r} = \rho\mathbf{N} + c\mathbf{B}$ with respect to s and impose that the centre $\mathbf{C}$ is fixed (to third order), i.e. $\mathbf{C}' = \mathbf{0}$ in the osculating sense. Differentiating:

$$-\mathbf{T} = \rho'\mathbf{N} + \rho\mathbf{N}' + c'\mathbf{B} + c\mathbf{B}'.$$

Substituting Frenet–Serret formulas:

$$-\mathbf{T} = \rho'\mathbf{N} + \rho(-\kappa\mathbf{T} + \tau\mathbf{B}) + c'\mathbf{B} + c(-\tau\mathbf{N}).$$

Collecting components:

$$\mathbf{T}\text{-component: } -1 = -\rho\kappa = -1. \quad \checkmark$$

$$\mathbf{N}\text{-component: } 0 = \rho' - c\tau \implies c\tau = \rho' \implies c = \frac{\rho'}{\tau} = \rho'\sigma. \quad (3)$$

$$\mathbf{B}\text{-component: } 0 = \rho\tau + c'.$$

The $\mathbf{T}$ -component is automatically satisfied (confirming consistency with Step 3), and equation (3) uniquely determines c .

Step 5: Centre and radius.

Substituting (1), (2), (3):

$$\boxed{\mathbf{C} = \mathbf{r} + \rho\mathbf{N} + \frac{\rho'}{\tau}\mathbf{B}}$$

The radius is:

$$R^2 = |\mathbf{C} - \mathbf{r}|^2 = b^2 + c^2 = \rho^2 + \left(\frac{\rho'}{\tau}\right)^2,$$

$$\boxed{R = \sqrt{\rho^2 + \left(\frac{\rho'}{\tau}\right)^2}}$$

□

Explicit Form in Terms of κ and τ

Since $\rho = 1/\kappa$, we have $\rho' = -\kappa'/\kappa^2$. Therefore:

$$\frac{\rho'}{\tau} = \frac{-\kappa'}{\kappa^2\tau},$$

and the centre and radius become:

$$\mathbf{C} = \mathbf{r} + \frac{1}{\kappa}\mathbf{N} - \frac{\kappa'}{\kappa^2\tau}\mathbf{B},$$

$$R = \sqrt{\frac{1}{\kappa^2} + \frac{\kappa'^2}{\kappa^4\tau^2}}.$$

Geometric Interpretation

1. The centre $\mathbf{C}$ lies in the **normal plane** (perpendicular to $\mathbf{T}$) through $\mathbf{r}$, since the $\mathbf{T}$ -component is zero.
2. The $\mathbf{N}$ -component $\rho = 1/\kappa$ is exactly the radius of curvature, placing $\mathbf{C}$ at distance ρ from the curve in the $\mathbf{N}$ direction — the same as the centre of the osculating circle.
3. The $\mathbf{B}$ -component ρ'/τ measures how far the sphere's centre is displaced *out of* the osculating plane, governed by how quickly the curvature is changing (ρ') relative to the torsion (τ).
4. When $\kappa' = 0$ (constant curvature, e.g. a circle or circular helix), $\rho' = 0$ and the centre reduces to $\mathbf{r} + \rho\mathbf{N}$, which is just the centre of the osculating circle — as expected.
5. When $\tau = 0$ (planar curve), the formula breaks down ($c \rightarrow \infty$), reflecting the fact that for a planar curve the osculating sphere degenerates to the osculating circle (a sphere of infinite radius in the limit).

Summary

$$\text{Centre:} \quad \mathbf{C} = \mathbf{r} + \underbrace{\rho}_{\text{along } \mathbf{N}} \mathbf{N} + \underbrace{\frac{\rho'}{\tau}}_{\text{along } \mathbf{B}} \mathbf{B}$$

$$\text{Radius:} \quad R = \sqrt{\rho^2 + \left(\frac{\rho'}{\tau}\right)^2}$$

Q4-A: Provide examples of surfaces in $\mathbb{R}^3$ and explain their classification into types such as planes, spheres, and cylinders.

1 Introduction

A **surface** in $\mathbb{R}^3$ is a two-dimensional set of points satisfying an equation of the form

$$F(x, y, z) = 0,$$

where $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a given function. Surfaces are classified primarily by the degree of F and by their geometric properties (flatness, curvature, symmetry). The most important family is that of *quadric surfaces*, defined by second-degree polynomials in x, y, z .

2 Linear Surfaces: Planes

Definition 1 (Plane). A **plane** in $\mathbb{R}^3$ is the set of points (x, y, z) satisfying

$$ax + by + cz = d,$$

where $a, b, c, d \in \mathbb{R}$ and $(a, b, c) \neq (0, 0, 0)$. The vector $\mathbf{n} = (a, b, c)$ is called the *normal vector* to the plane.

Planes are the simplest surfaces: they have zero curvature everywhere and divide $\mathbb{R}^3$ into two half-spaces.

Example 2 (Coordinate planes).

- xy -plane: $z = 0$.
- yz -plane: $x = 0$.
- xz -plane: $y = 0$.

Example 3 (General plane). The plane passing through the origin with normal $\mathbf{n} = (1, 2, 3)$ is

$$x + 2y + 3z = 0.$$

3 Quadric Surfaces

Definition 4 (Quadric surface). A **quadric surface** is the solution set of a general second-degree equation

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0,$$

where the coefficients $A, B, \dots, J \in \mathbb{R}$ are not all zero. By a suitable rotation and translation of axes, every non-degenerate quadric can be reduced to one of the standard forms listed below.

3.1 Sphere

Definition 5 (Sphere). A **sphere** with centre $C = (x_0, y_0, z_0)$ and radius $r > 0$ is

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = r^2.$$

It is a quadric with equal positive coefficients on all three squared terms.

Key properties:

- Every cross-section by a plane is a circle (or a point, or empty).
- The sphere has constant positive Gaussian curvature $K = 1/r^2$.

Example 6. The sphere centred at the origin with radius 2:

$$x^2 + y^2 + z^2 = 4.$$

3.2 Ellipsoid

Definition 7 (Ellipsoid). An **ellipsoid** with semi-axes $a, b, c > 0$ has the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

When $a = b = c = r$, this reduces to a sphere of radius r .

Example 8.

$$\frac{x^2}{9} + \frac{y^2}{4} + \frac{z^2}{1} = 1$$

is an ellipsoid with semi-axes $a = 3, b = 2, c = 1$.

3.3 Cylinder

Definition 9 (Cylinder). A **(right circular) cylinder** aligned along the z -axis with radius $r > 0$ is

$$x^2 + y^2 = r^2.$$

More generally, any equation that is *independent of one variable* defines a cylindrical surface: the curve in the remaining two variables is *extruded* infinitely along the missing axis.

Key property: One eigenvalue of the associated matrix is zero, reflecting the lack of curvature along the axis direction.

Example 10 (Circular cylinder). $x^2 + y^2 = 9$ — radius-3 cylinder along the z -axis.

Example 11 (Parabolic cylinder). $y = x^2$ — a parabolic cylinder extending along the z -axis.

Example 12 (Elliptic cylinder). $\frac{x^2}{4} + \frac{y^2}{9} = 1$ — elliptic cross-section, extends along the z -axis.

3.4 Cone

Definition 13 (Elliptic cone). An **elliptic cone** with vertex at the origin is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0.$$

For $a = b$ this becomes a *right circular cone*.

Example 14. $x^2 + y^2 - z^2 = 0$, or equivalently $z = \pm\sqrt{x^2 + y^2}$, is a right circular cone with half-angle 45° .

3.5 Paraboloids

Definition 15 (Elliptic paraboloid). An **elliptic paraboloid** has the equation

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}.$$

Definition 16 (Hyperbolic paraboloid (saddle surface)). A **hyperbolic paraboloid** (saddle) has the equation

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}.$$

Example 17. $z = x^2 + y^2$ is an elliptic paraboloid (a bowl opening upward).

Example 18. $z = x^2 - y^2$ is a hyperbolic paraboloid (a saddle at the origin).

3.6 Hyperboloids

Definition 19 (Hyperboloid of one sheet).

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1.$$

This is a *connected* surface (one piece).

Definition 20 (Hyperboloid of two sheets).

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1, \quad \text{equivalently} \quad \frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

This is a *disconnected* surface (two separate sheets).

Example 21. $x^2 + y^2 - z^2 = 1$: hyperboloid of one sheet (hourglass-like, connected).

Example 22. $z^2 - x^2 - y^2 = 1$: hyperboloid of two sheets (two separate bowls).

4 Classification Summary

Surface	Standard Equation	Key Feature
Plane	$ax + by + cz = d$	Degree 1; flat
Sphere	$x^2 + y^2 + z^2 = r^2$	Equal coefficients, $K > 0$
Ellipsoid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	Generalised sphere
Circular Cylinder	$x^2 + y^2 = r^2$	Missing variable (z)
Elliptic Cylinder	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	Missing variable, elliptic cross-section
Parabolic Cylinder	$y = x^2$	Missing variable, parabolic cross-section
Cone	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$	Vertex at origin
Elliptic Paraboloid	$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$	Opens in one direction
Hyperbolic Paraboloid	$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$	Saddle-shaped
Hyperboloid sheet)	(1) $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	Connected; waist circle
Hyperboloid sheets)	(2) $\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	Two disconnected pieces

5 Classification Criteria

Given a quadric $F(x, y, z) = 0$, the standard classification proceeds as follows.

- Degree of equation.** If F is linear ($\deg F = 1$), the surface is a *plane*.
- Signs of squared-term coefficients.** After reducing to standard form, count the positive (p) and negative (q) squared terms:
 - $p = 3, q = 0$: ellipsoid (sphere if all coefficients equal).
 - $p = 2, q = 1$: hyperboloid of one sheet or cone ($= 0$ RHS).
 - $p = 1, q = 2$: hyperboloid of two sheets.
 - $p = 2, q = 0$ with one linear term: elliptic/circular paraboloid.
 - $p = 1, q = 1$ with one linear term: hyperbolic paraboloid.
- Missing variable.** If F is independent of one coordinate (say z), the surface is a *cylinder* whose cross-section is determined by the two-variable curve $F(x, y) = 0$.

4. **Right-hand side.** For surfaces of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = k$: $k > 0$ gives one sheet, $k = 0$ gives a cone, $k < 0$ gives two sheets.

6 Conclusion

Surfaces in $\mathbb{R}^3$ range from the simplest flat *plane* (degree 1) to the rich family of *quadric surfaces* (degree 2). The key classification criteria are: the degree of the defining polynomial, the signs of the squared-variable coefficients, whether a variable is absent (cylindrical surface), and the value of the constant term. Understanding these distinctions is fundamental in multivariable calculus, differential geometry, and linear algebra.

Q5-A: Explain the second fundamental form of a surface. Derive the expression for the normal curvature and provide a geometrical interpretation.

1 The Second Fundamental Form

1.1 Motivation

The first fundamental form tells us nothing about how the surface bends in $\mathbb{R}^3$. To capture bending, we need to compare the surface to its tangent plane by measuring how much the unit normal $\hat{\mathbf{n}}$ changes as we move along the surface.

1.2 Derivation of the Coefficients

Since $\hat{\mathbf{n}} \cdot \mathbf{r}_u = 0$ and $\hat{\mathbf{n}} \cdot \mathbf{r}_v = 0$ (the normal is perpendicular to all tangent vectors), differentiating the first equation with respect to u gives

$$\hat{\mathbf{n}}_u \cdot \mathbf{r}_u + \hat{\mathbf{n}} \cdot \mathbf{r}_{uu} = 0 \implies \hat{\mathbf{n}} \cdot \mathbf{r}_{uu} = -\hat{\mathbf{n}}_u \cdot \mathbf{r}_u.$$

This shows that the second-order variation of $\mathbf{r}$ projected onto $\hat{\mathbf{n}}$ equals (up to sign) the variation of $\hat{\mathbf{n}}$ projected back onto the tangent plane.

Definition 1 (Second Fundamental Form). The **second fundamental form** is the symmetric bilinear form

$$\text{II} = L du^2 + 2M du dv + N dv^2,$$

where the coefficients are defined by

$$L = \mathbf{r}_{uu} \cdot \hat{\mathbf{n}}, \quad M = \mathbf{r}_{uv} \cdot \hat{\mathbf{n}}, \quad N = \mathbf{r}_{vv} \cdot \hat{\mathbf{n}}.$$

Equivalently, using $|\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{EG - F^2}$,

$$L = \frac{(\mathbf{r}_{uu}, \mathbf{r}_u, \mathbf{r}_v)}{\sqrt{EG - F^2}}, \quad M = \frac{(\mathbf{r}_{uv}, \mathbf{r}_u, \mathbf{r}_v)}{\sqrt{EG - F^2}}, \quad N = \frac{(\mathbf{r}_{vv}, \mathbf{r}_u, \mathbf{r}_v)}{\sqrt{EG - F^2}},$$

where $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ denotes the scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$.

Remark. Unlike E, F, G (which depend only on the metric), the coefficients L, M, N depend on the *embedding* of the surface in $\mathbb{R}^3$ and on the choice of normal orientation.

2 Derivation of Normal Curvature

2.1 Normal Sections

Definition 2 (Normal Section). Let P be a point on the surface and $\mathbf{t}$ a unit tangent vector at P . The **normal section** in the direction $\mathbf{t}$ is the curve γ obtained by intersecting the surface with the plane spanned by $\mathbf{t}$ and $\hat{\mathbf{n}}(P)$.

2.2 Step-by-Step Derivation

Let $\gamma(t) = \mathbf{r}(u(t), v(t))$ be a smooth curve on the surface passing through $P = \mathbf{r}(u_0, v_0)$ at $t = 0$, with unit speed: $|\dot{\gamma}| = 1$.

Step 1: First derivative.

$$\dot{\gamma} = \mathbf{r}_u \dot{u} + \mathbf{r}_v \dot{v}.$$

Step 2: Second derivative.

$$\ddot{\gamma} = \mathbf{r}_{uu} \dot{u}^2 + 2\mathbf{r}_{uv} \dot{u} \dot{v} + \mathbf{r}_{vv} \dot{v}^2 + \mathbf{r}_u \ddot{u} + \mathbf{r}_v \ddot{v}.$$

Step 3: Project onto the normal. Since $\mathbf{r}_u$ and $\mathbf{r}_v$ are tangent vectors, $\hat{\mathbf{n}} \cdot \mathbf{r}_u = \hat{\mathbf{n}} \cdot \mathbf{r}_v = 0$. Therefore

$$\ddot{\gamma} \cdot \hat{\mathbf{n}} = L \dot{u}^2 + 2M \dot{u} \dot{v} + N \dot{v}^2.$$

Step 4: Identify the normal curvature. For a unit-speed curve, the curvature vector is $\ddot{\gamma}$. Its component along $\hat{\mathbf{n}}$ is the **normal curvature**:

$$\kappa_n = \ddot{\gamma} \cdot \hat{\mathbf{n}} = L \dot{u}^2 + 2M \dot{u} \dot{v} + N \dot{v}^2.$$

Step 5: Express as a ratio of fundamental forms. For an arbitrary (not necessarily unit-speed) curve with tangent $d\mathbf{r} = \mathbf{r}_u du + \mathbf{r}_v dv$, we have

$$I = E du^2 + 2F du dv + G dv^2 = |d\mathbf{r}|^2 > 0,$$

so the unit-speed condition normalises the denominator to 1. In general,

Theorem 3 (Normal Curvature Formula). *The normal curvature of a surface $\mathbf{r}(u, v)$ in the tangent direction $(du : dv)$ is*

$$\kappa_n = \frac{II}{I} = \frac{L du^2 + 2M du dv + N dv^2}{E du^2 + 2F du dv + G dv^2}.$$

For a unit tangent vector $\mathbf{t} = a\mathbf{r}_u + b\mathbf{r}_v$ with $Ea^2 + 2Fab + Gb^2 = 1$, this simplifies to

$$\kappa_n = La^2 + 2Mab + Nb^2.$$

3 Principal Curvatures, Gaussian and Mean Curvature

3.1 Principal Curvatures

The normal curvature κ_n varies with direction. Its extreme values are found by solving the eigenvalue problem for the **shape operator** (Weingarten map) $\mathcal{S}$, defined by $\mathcal{S}(\mathbf{t}) = -d\hat{\mathbf{n}}(\mathbf{t})$.

In matrix form relative to the basis $\{\mathbf{r}_u, \mathbf{r}_v\}$, the condition $\kappa_n = \lambda$ gives the characteristic equation:

$$\det \begin{pmatrix} L - \lambda E & M - \lambda F \\ M - \lambda F & N - \lambda G \end{pmatrix} = 0,$$

whose roots $\kappa_1 \leq \kappa_2$ are the **principal curvatures**. The corresponding directions are the **principal directions**.

3.2 Gaussian and Mean Curvature

Definition 4. The **Gaussian curvature** and **mean curvature** are

$$K = \kappa_1 \kappa_2 = \frac{LN - M^2}{EG - F^2}, \quad H = \frac{\kappa_1 + \kappa_2}{2} = \frac{EN - 2FM + GL}{2(EG - F^2)}.$$

Remark. By Gauss's *Theorema Egregium*, K is an *intrinsic* invariant: it depends only on E, F, G and their derivatives, not on L, M, N directly.

4 Geometric Interpretation

Geometric Interpretation. The second fundamental form and normal curvature have the following geometric meanings.

(i) Signed distance to the tangent plane. Let P be a point on the surface with tangent plane Π_P . For a nearby point $Q = \mathbf{r}(u_0 + \delta u, v_0 + \delta v)$, Taylor expansion gives

$$\overrightarrow{PQ} = \underbrace{(\mathbf{r}_u \delta u + \mathbf{r}_v \delta v)}_{\in \Pi_P} + \frac{1}{2}(\mathbf{r}_{uu} \delta u^2 + 2\mathbf{r}_{uv} \delta u \delta v + \mathbf{r}_{vv} \delta v^2) + O(\delta^3).$$

The signed distance from Q to the tangent plane is

$$d(Q, \Pi_P) = \overrightarrow{PQ} \cdot \hat{\mathbf{n}} = \frac{1}{2}(L \delta u^2 + 2M \delta u \delta v + N \delta v^2) + O(\delta^3) = \frac{1}{2} \Pi(\delta u, \delta v) + O(\delta^3).$$

Therefore, *the second fundamental form measures, to second order, how much the surface deviates from its tangent plane.*

(ii) Curvature of curves on the surface. For any curve γ on the surface, its curvature vector decomposes as

$$\ddot{\gamma} = \kappa_n \hat{\mathbf{n}} + \kappa_g (\hat{\mathbf{n}} \times \dot{\gamma}),$$

where κ_g is the **geodesic curvature** (tangential bending) and κ_n is the normal curvature. All curves through P in the same tangent direction share the same κ_n (this is *Meusnier's theorem*).

(iii) Sign of κ_n and surface shape. At a point P :

- $K > 0$ (κ_1 and κ_2 same sign): surface is *elliptic* (bowl-shaped); the surface lies on one side of the tangent plane nearby.
- $K < 0$ (κ_1 and κ_2 opposite signs): surface is *hyperbolic* (saddle-shaped); the surface crosses the tangent plane in every neighbourhood.
- $K = 0$ (at least one principal curvature zero): surface is *parabolic* (e.g. a cylinder) or *flat* (a plane).

(iv) **Euler's formula for normal curvature.** If the coordinate directions are chosen to coincide with the principal directions, then $M = 0$ (so $F = 0$ and $E = G = 1$ for a unit-speed frame), and for a tangent direction making angle θ with the first principal direction,

$$\kappa_n(\theta) = \kappa_1 \cos^2 \theta + \kappa_2 \sin^2 \theta.$$

This is **Euler's formula**: normal curvature varies smoothly between the two extreme values κ_1 and κ_2 .